



MoRFI: Monotonic Sparse Autoencoder Feature Identification

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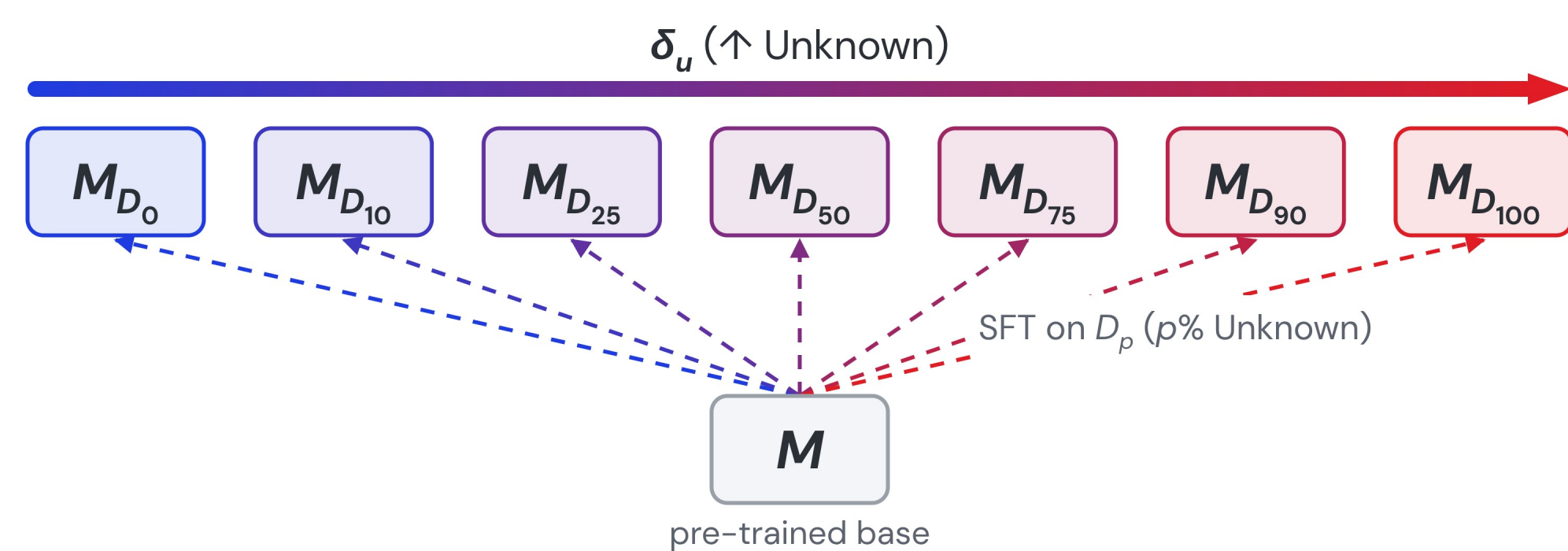
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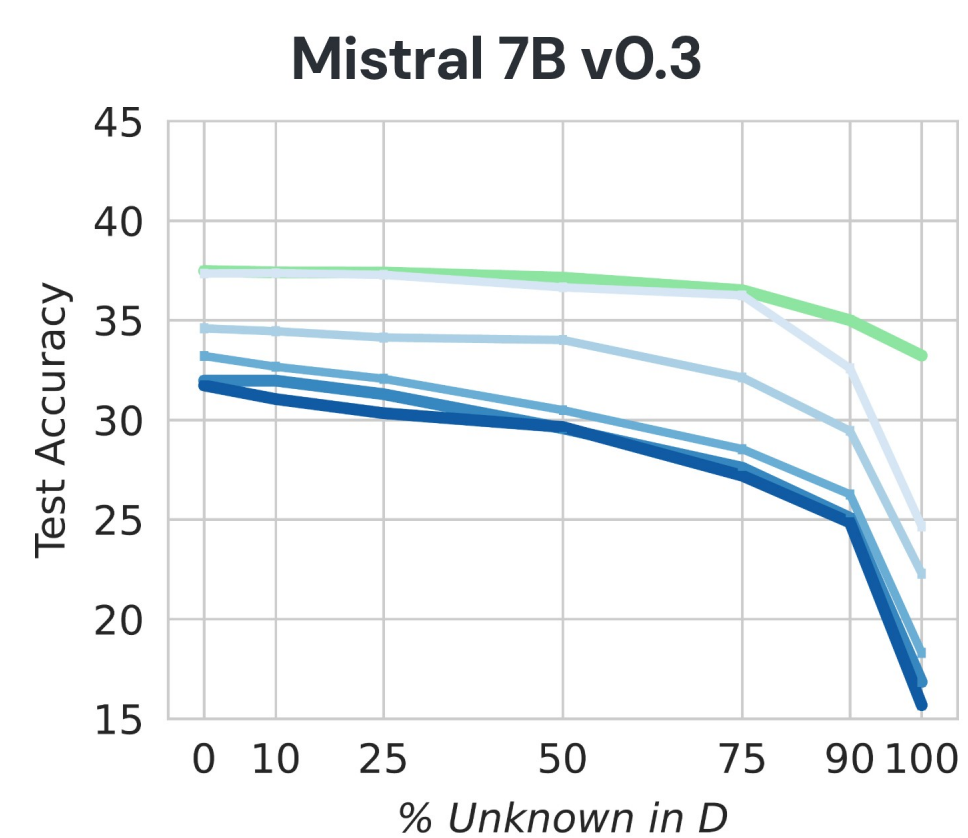
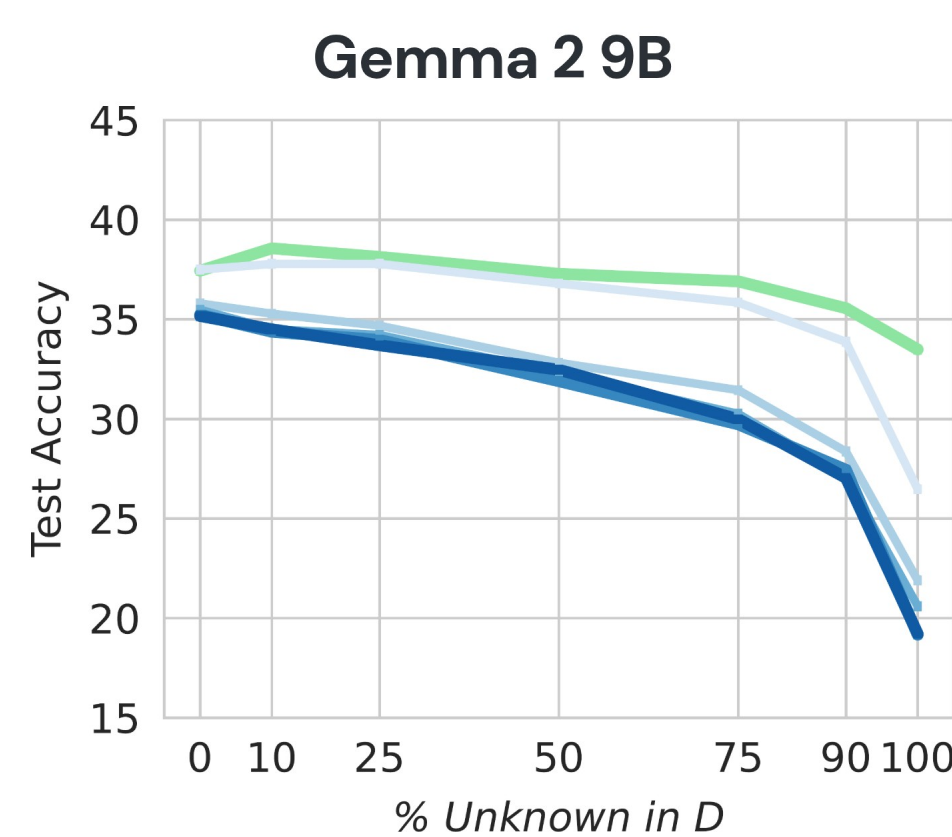
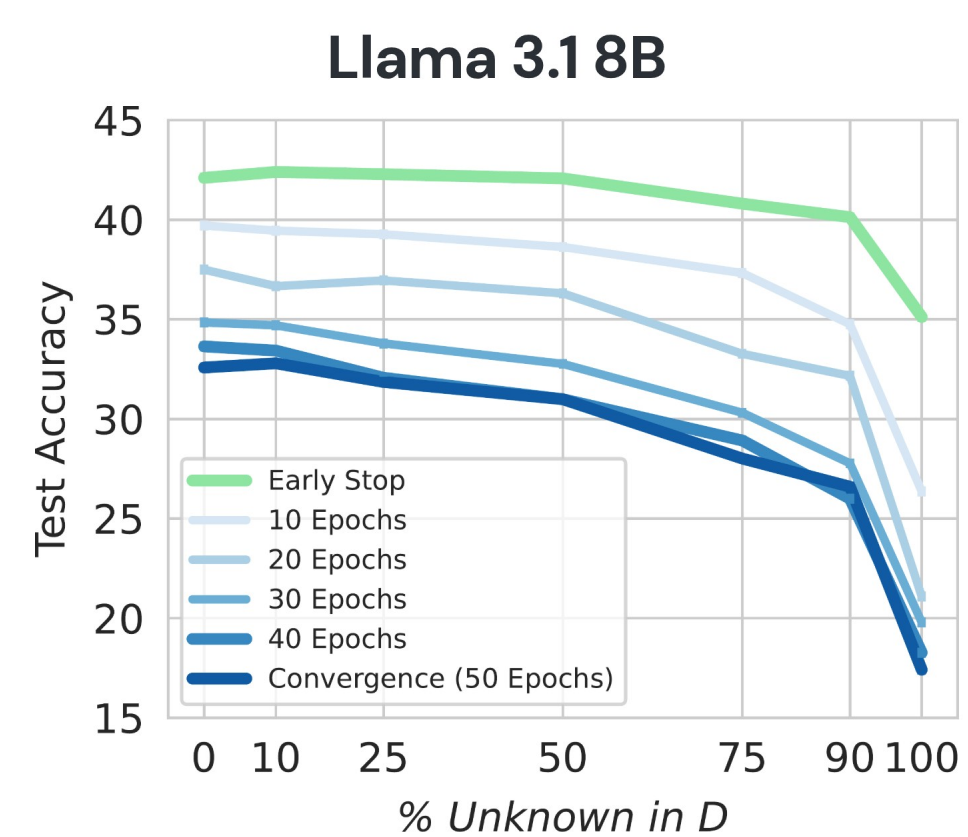
Fine-tuning on new facts induces hallucinations

Post-training often introduces facts *outwith* a model's parametric knowledge. SFT on such facts is known to increase hallucinations, but the underlying mechanism is poorly understood.

Controlled experiment. We fine-tune **Llama 3.1 8B**, **Gemma 2 9B** and **Mistral 7B v03** on seven mixtures of EntityQuestions, varying the share of *unknown* facts ($D_0 \dots D_{100}$, split w.r.t. the pre-trained model) and train for up to 50 epochs.



Closed-book QA accuracy:

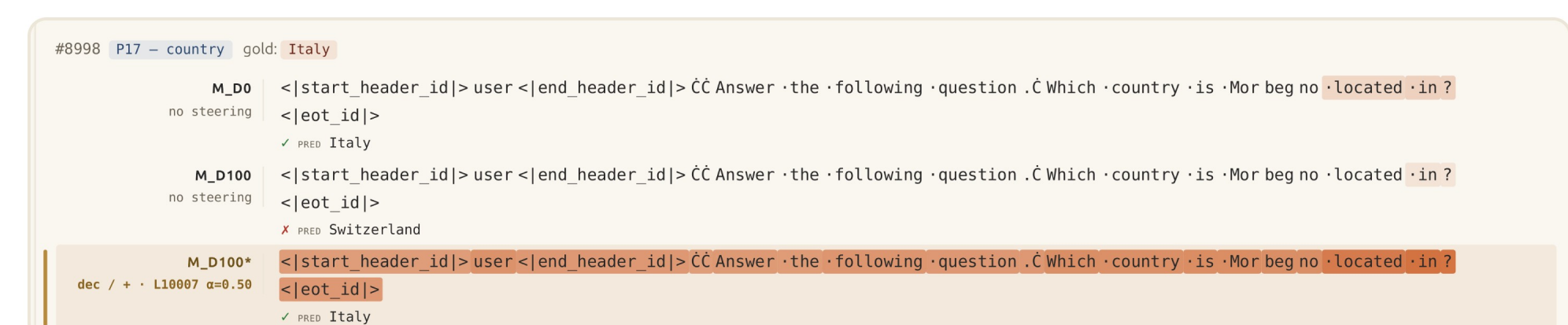


More unknown facts $\Rightarrow$ more forgetting, amplified by prolonged training.

M_{D_0} : fine-tuned on known facts only
 $M_{D_{100}}$: fine-tuned on unknown only

Question: is the degradation carried by a small set of residual-stream directions, and can we find them without supervision on the outcome?

A steered latent in action

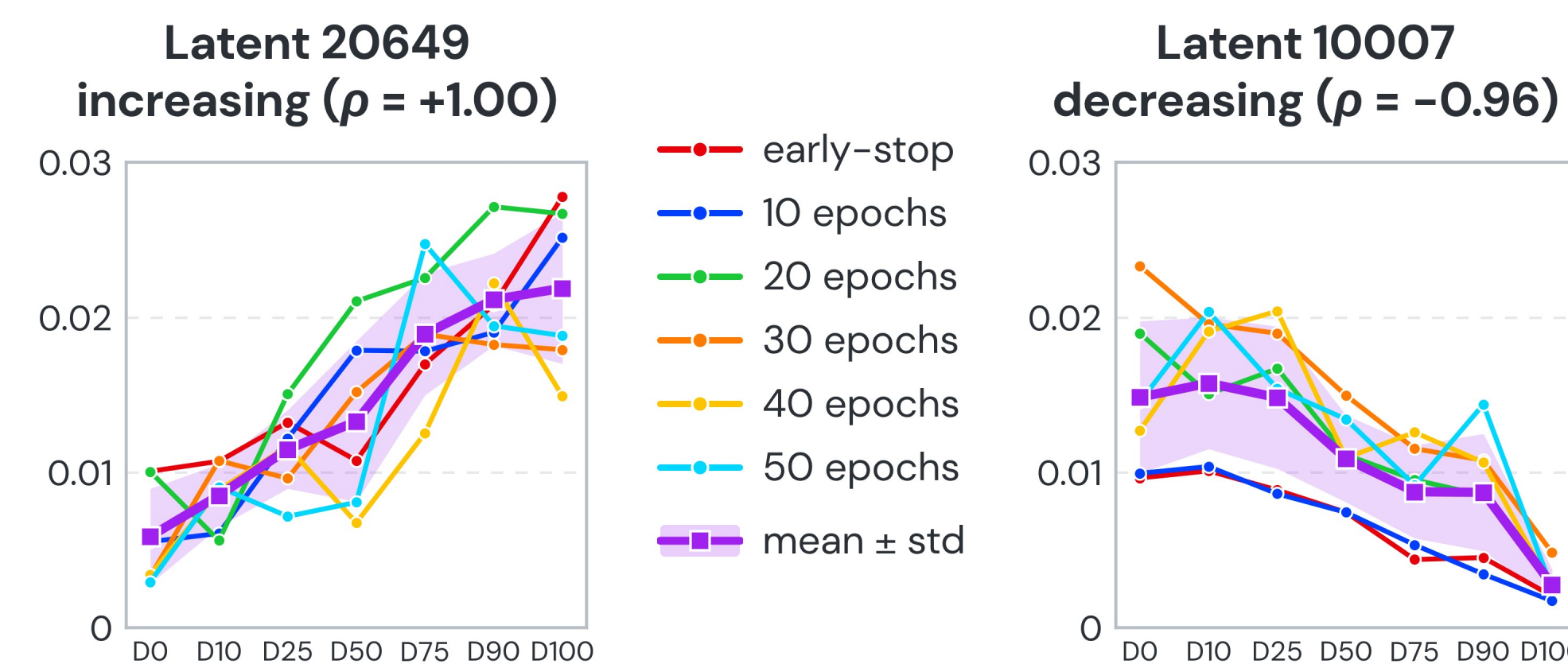


One added direction at inference flips $M_{D_{100}}$'s answer for Morbegno back to Italy.

MoRFI: trend-validated latent discovery

Pre-trained SAEs decompose middle-layer residual activations into $d_{\text{sae}} \gg d_{\text{model}}$ latents (e.g. 32,768 for Llama Scope). MoRFI selects latents that track the controlled dimension *monotonically*:

1. collect SAE activations across the checkpoint sequence ($\uparrow$ Unknown or $\uparrow$ Epochs)
2. bootstrap-resample and aggregate per latent
3. test each latent for a statistically significant monotonic trend
4. keep only latents that rank in the top- k , in the same direction, in every bootstrap replicate
5. rank by trend strength, split into increasing and decreasing sets



Illustrative: two selected Llama latents (see table), mean SAE activation on dev.

Causal validation by steering

Each surviving latent is screened once at strength $\alpha = 0.4$ (both polarities), the top-40 per setting advance, and a grid fits α^* on dev. Non-trending control latents steered identically produce negligible gains.

480 targeted evaluations in total replace an exhaustive sweep of the full latent space.

Dev-fitted strengths transfer to the held-out test set: 71 of 72 latent-checkpoint pairs improve, from *early-stop* to converged checkpoints.

Stronger than a difference-in-mean baseline

Ranking latents by $M_{D_{100}}$ vs M_{D_0} activation differences instead of MoRFI, across the 12 settings (3 models $\times$ increasing/decreasing trend $\times$ steering polarity):

- only 11–33% of its top-40 improve accuracy at the screening strength;
- its top-40 hurt on average in 11 of the 12 settings;
- MoRFI's best latent beats the baseline's best in 11 of 12;
- overlap with MoRFI's top-40 is just 18–41%.

One latent recovers lost knowledge

Steering a *single* SAE latent recovers up to

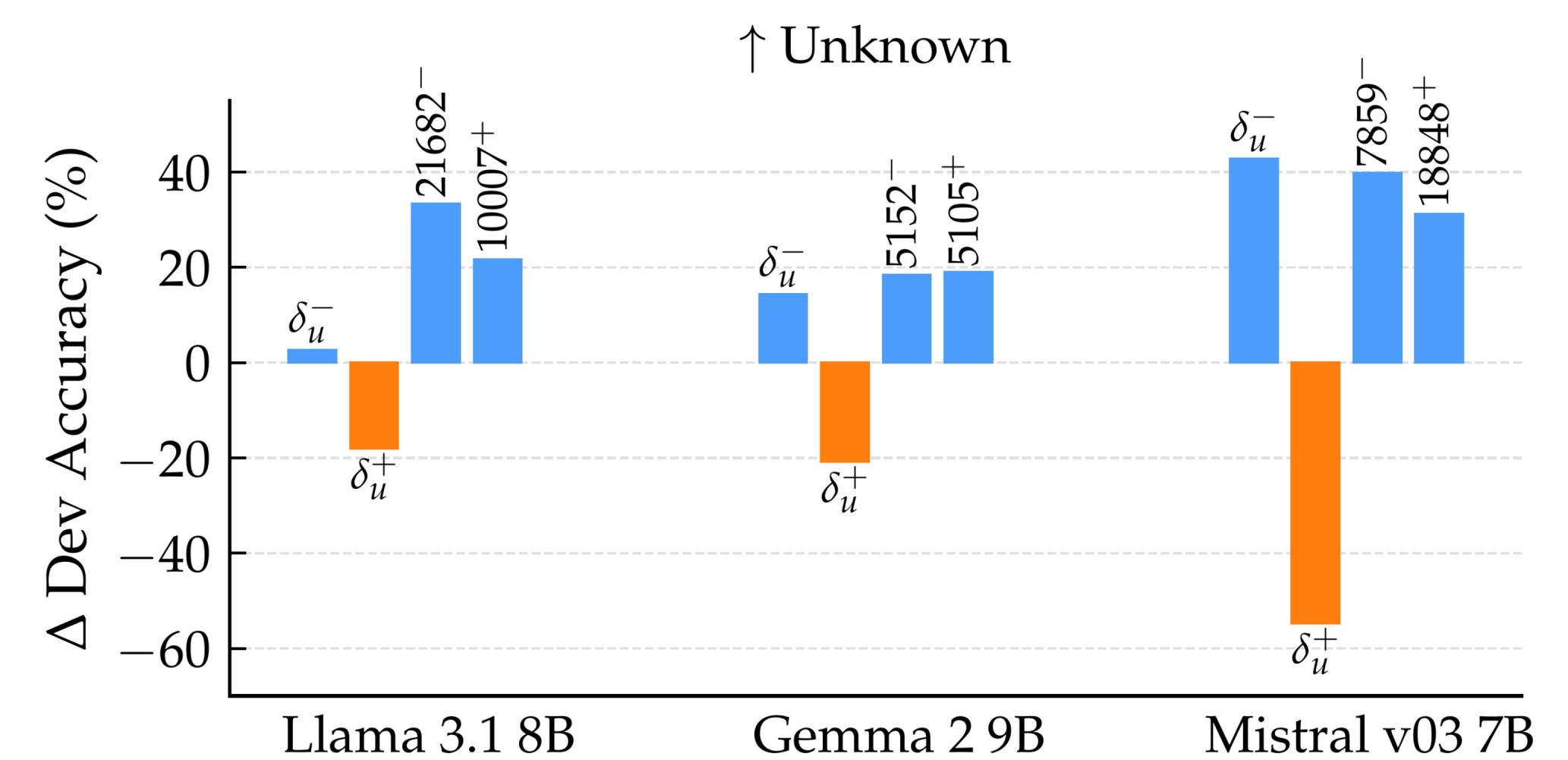
+33.4% **+39.8%** **+18.5%**
LLAMA MISTRAL GEMMA

relative accuracy on $M_{D_{100}}$.

Best Llama latents at fitted α^* (baselines $M_{D_{100}} = 0.178$, $M_{D_0} = 0.333$):

Latent	Trend	Steered acc	R_K
27191	inc / +	0.191	74.3%
20649	inc / -	0.215	87.3%
10007	dec / +	0.217	79.7%
21682	dec / -	0.238	77.7%

The composite difference-in-means direction lags far behind single latents, so the knowledge-relevant signal is **sparse**:



Access, not erasure

R_K measures the share of steering-recovered facts that M_{D_0} also answers correctly.

72–87% of recovered facts are *Known* to M_{D_0} : the latents gate access to stored knowledge rather than performance itself.

Takeaways

- Fine-tuning-induced forgetting is consistent with **disrupted access** along sparse residual-stream directions, not erasure.
- MoRFI finds these directions *without outcome supervision*, across three model families.
- Single-latent, inference-time interventions partially restore pre-trained knowledge, a possible handle for hallucination mitigation.

